

COMPREHENSIVE EXERCISES

I. NumPy Library

Exercise 1: Gold Price Trend Analysis

- **Dataset:** Gold Prices (1970-2021)
 - Description: Contains date columns and corresponding gold prices.
 - URL: https://github.com/MYoussef885/Gold_Price_Prediction/blob/main/gold_price_data.csv
- **Requirements:**
 - Download the data and extract the "GLD" column (gold price) as a NumPy array.
 - Calculate the daily return rate using the formula:
$$r_t = \frac{Price_t - Price_{t-1}}{Price_{t-1}}$$
 - Find the mean and standard deviation of the return rates. Identify days with high volatility (absolute return rate > 3%).
- **Output:** Write the statistical indices (mean, std) and the list of high-volatility days to a file named `gold_analysis_report.txt`.

Exercise 2: Air Quality Monitoring

- **Dataset:** Air Quality Data in India
 - Description: AQI index and concentrations of pollutants (PM2.5, PM10, NO2...).
 - URL: https://raw.githubusercontent.com/mimansha98/AQI-Prediction-before-and-after-Lockdown-of-COVID-19-in-India/refs/heads/main/city_day.csv
- **Requirements:**
 - Read the data and handle missing values (NaN) by replacing them with the mean of the respective column.
 - Calculate the average air quality index for each pollutant column.
 - Perform "Binning": Classify rows into 3 levels: "Good" (AQI < 50), "Moderate" (50-100), and "Poor" (AQI > 100) using `np.digitize` or `np.where`.
- **Output:** Export the processed data array with the new classification column to a file named `air_quality_processed.csv`.

Exercise 3: Student Performance Analysis

- **Dataset:** Student Performance Dataset
 - Description: Student scores and demographic factors (use file **student-mat.csv**). Note: Delimited by ;.
 - URL: <https://www.kaggle.com/dskagglemt/student-performance-data-set>
- **Requirements:**
 - Read columns G1, G2, and G3 (the last 3 columns).
 - Calculate the final term score based on weights:
Final = G1 x 0.25 + G2 x 0.25 + G3 x 0.5
 - Use array comparison to find the percentage of students with a *Final* ≥ 10.

- Calculate the Covariance Matrix between G1, G2, and G3 to observe correlations.
- **Output:** Save the covariance matrix and the student pass rate to a file named student_stats.txt.

Exercise 4: Heart Rate Signal Processing

- **Dataset:**
 - ECG Sequence Data (Use any floating-point data representing a time-series signal).
 - URL: <https://raw.githubusercontent.com/jbrownlee/Datasets/master/pollution.csv>
- **Requirements:**
 - Simulate a noisy heart rate signal by adding Gaussian Noise to the original data using `np.random.normal`.
 - Use the "Moving Average" technique with a sliding window (window size = 5) to smooth the signal (noise reduction).
 - Find local maxima in the signal to estimate the heart rate.
- **Output:** Save the smoothed signal array to a file named smooth_signal.csv.

Exercise 5: Sales Prediction using Linear Regression

- **Dataset:** Advertising Dataset
 - Description: Advertising expenditures on TV, Radio, Newspaper, and resulting Sales.
 - URL: <https://raw.githubusercontent.com/juSTmarkham/scikit-learn-videos/master/data/Advertising.csv>
 - **Requirements:**
 - Split the data into an independent variable X (TV column) and a dependent variable y (Sales column).
 - Apply the Least Squares method to find coefficients a and b for the linear equation:
- $$y = ax + b: a = \frac{\sum(x - \bar{x})(y - \bar{y})}{\sum(x - \bar{x})^2}$$
- Predict sales for a newly created array of advertising costs.
- **Output:** Save coefficients a, b and the predicted values to a file named sales_model_results.txt.

II. Pandas Library

Exercise 6: Netflix Content Analysis

- **Dataset:**
 - Netflix Movies and TV Shows
 - URL: https://raw.githubusercontent.com/prasertcbs/basic-dataset/refs/heads/master/netflix_titles.csv
- **Requirements:**
 - Load the data and handle missing values in the **country**, **cast**, and **director** columns by filling them with "Unknown".
 - Filter for movies (Type is 'Movie') produced after the year 2015.
 - Calculate the count of content per country (Note: one entry may belong to multiple countries; string splitting is required).
 - Create a summary table of the number of Movies and TV Shows added by month (using the **date_added** column).
- **Output:** Export the monthly trend summary table to a file named `netflix_monthly_trends.csv`.

Exercise 7: Global Retail Sales Analysis

- **Dataset:**
 - Online Retail Dataset
 - URL: <https://raw.githubusercontent.com/databricks/Spark-The-Definitive-Guide/refs/heads/master/data/retail-data/all/online-retail-dataset.csv>
- **Requirements:**
 - Remove canceled orders (Invoice numbers starting with 'C') and rows where **Quantity** or **UnitPrice** is less than or equal to 0.
 - Create a new column: **TotalSales = Quantity * UnitPrice**.
 - Use the *groupby* function to calculate the total revenue and number of orders per **Country**.
 - Identify the Top 5 customers (**CustomerID**) with the highest total spending.
- **Output:** Save the Top 5 customers and revenue by country to an Excel file named `retail_summary.xlsx`.

Exercise 8: Weather Data Merging and Analysis

- **Dataset:**
 - Historical Weather Data
 - URL: <https://raw.githubusercontent.com/jbrownlee/Datasets/master/daily-min-temperatures.csv>
- **Requirements:**
 - Convert the date column to *datetime* format.
 - Extract **Year**, **Month**, and **Day** into new separate columns.
 - Calculate the average, maximum, and minimum temperatures for each month of the year.
 - Use the *resample* method to calculate the average temperature on a quarterly basis.

- **Output:** Save the quarterly statistical results to a text file named `weather_quarterly_report.txt`.

Exercise 9: Stock Market Pivot Analysis

- **Dataset:**
 - S&P 500 Companies Financials
 - URL: <https://github.com/datasets/s-and-p-500-companies-financials/blob/main/data/constituents-financials.csv>
- **Requirements:**
 - Filter the data for companies in sectors such as 'Information Technology' and 'Health Care'.
 - Calculate the average **Price/Earnings** ratio for each sector.
 - Use *pivot_table* to compare the average **Market Cap** across different sectors.
 - Identify companies with an **EBITDA** higher than their respective sector's average.
- **Output:** Export the sector market cap comparison table to a file named `sector_comparison.csv`.

Exercise 10: Flight Delay Analysis

- **Dataset:**
 - Flight Delays Data
 - URL: <https://raw.githubusercontent.com/vaibhavwalvekar/NYC-Flights-2013-Dataset-Analysis/refs/heads/master/flights.csv>
- **Requirements:**
 - Handle **NaN** values in the **dep_delay** and **arr_delay** columns by removing those rows.
 - Calculate the total average delay time for each airline (**carrier**).
 - Find the peak hour of the day with the highest percentage of delayed flights.
 - Merge this table with the airline information table (from: Airlines Info) to display full airline names instead of codes.
- **Output:** Save the list of airlines and their average delay times to a file named `airline_delay_analysis.csv`.

III. Matplotlib Library

Exercise 11: Stock Price Trend Analysis

- **Dataset:**
 - S&P 500 Stock Prices
 - URL: <https://github.com/mwitiderrick/stockprice/blob/master/NSE-TATAGLOBAL.csv>
- **Requirements:**
 - Plot a Line graph representing the **Close** price over time.
 - Add a 30-day Moving Average line with a different color for comparison.
 - Annotate the highest and lowest price points on the graph.
 - Customize the X-axis to display dates clearly (rotate labels by 45 degrees).
- **Output:** Save the plot to a file named stock_trend_analysis.png.

Exercise 12: Air Quality Index Distribution

- **Dataset:**
 - Air Quality Data
 - URL: https://raw.githubusercontent.com/mimansha98/AQI-Prediction-before-and-after-Lockdown-of-COVID-19-in-India/refs/heads/main/city_day.csv
- **Requirements:**
 - Use a Histogram to represent the distribution of the AQI index.
 - Create a Box plot to compare the concentrations of PM2.5, PM10, and NO2 on the same coordinate axis.
 - Use a Colormap to distinguish air quality zones (e.g., green for low AQI, red for high AQI).
- **Output:** Save the combination of plots to a file named air_quality_distribution.pdf.

Exercise 13. Correlation Between Sales and Advertising

- **Dataset:**
 - Advertising Dataset
 - URL: <https://raw.githubusercontent.com/justmarkham/scikit-learn-videos/master/data/Advertising.csv>
- **Requirements:**
 - Create a Scatter plot between TV advertising costs and **Sales**.
 - Use the size of the points (Size) to represent Radio advertising costs.
 - Draw a linear regression line (Best-fit line) through the scatter points.
 - Add a title, axis labels, and a grid to the plot.
- **Output:** Save the plot to a file named advertising_correlation.png.

Exercise 14: Netflix Content Composition Comparison

- **Dataset:**
 - Netflix Titles

- URL: https://raw.githubusercontent.com/prasertcbs/basic-dataset/refs/heads/master/netflix_titles.csv
- **Requirements:**
 - Draw a Pie chart showing the percentage ratio between 'Movie' and 'TV Show'.
 - Create a Stacked Bar Chart showing the number of contents added over the years (select the most recent 5 years).
 - Use Netflix's signature color palette (Red and Black) to style the charts.
- **Output:** Save the charts to a file named netflix_composition.svg.

Exercise 15: Grouped Academic Performance Analysis

- **Dataset:**
 - Student Performance (Use the student-mat.csv file).
 - URL: <https://www.kaggle.com/datasets/dskagglemt/student-performance-data-set>
- **Requirements:**
 - Create a Grouped Bar Chart comparing average final scores (**G3**) between males (*M*) and females (*F*) across different family address types (**address**: Urban vs Rural).
 - Add Error bars representing the standard deviation of the scores.
 - Create a Subplot next to it to draw a density plot (KDE or Histogram) of the total scores.
- **Output:** Save the results to a file named student_performance_comparison.png.